

Key substitution attacks on TSO et al.'s short signature scheme

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SUMMARY

Recently, TSO et al. proposed an efficient pairing based short signature scheme at ITNG'09. In this letter, we show that their short signature scheme is susceptible to key substitution attacks.

Key words: key substitution attack, short signature scheme.

1 Introduction

In 1984, Goldwasser, Micali and Rivest (GMR) introduced the security notion of existential unforgeability against adaptive chosen-message attacks for signature schemes in the single-user setting [5].

In 2004, Menezes and Smart pointed out that the GMR security model is not sufficient in a multi-user setting, and they proposed a new type of attacks on digital signature schemes which they called key substitution (KS) attacks [7]. It is well known that one of the functions which signature schemes have is non-repudiation. As is pointed out by Kobitz and Menezes [6], a serious practical danger of KS attacks is that they'll undermine non-repudiation. AKS attack works as follows: given a legitimate user U 's public key and U 's signature σ on some message m , an adversary $\mathcal{A}$ tries to produce a new public key different from U 's such that σ is also a valid

signature on the same message m under the new public key $\mathcal{A}$ produced. In a KS attack, if the adversary can reveal the private key corresponding to the public key he chooses, we call this a weak-key substitution (WKS) attack, otherwise, a strong-key substitution (SKS) attack. Many signature schemes are susceptible to key substitution attacks: [1], [2], [4], [7], [9], [10].

Furthermore, in 2006, when considering KS attacks, Bohli et al. [2] pointed out that malicious signers (which can provide their secret informations even their private keys to an adversary) also should be taken into account.

In 2001, Boneh et al. [3] proposed the first pairing based short signature scheme which is also known as the BLS scheme. As noted in [3], short signature schemes are useful in low-bandwidth communication environments or in environments where a human is asked to manually key in the signature. At PKC'04, Zhang et al. [12] proposed an efficient pairing based short signature scheme which we call the ZSS scheme. Recently, at ITNG'09, TSO et al. [11] proposed a new pairing based short signature scheme (referred to as the TOO scheme below) which is more efficient than both the BLS scheme and the ZSS scheme.

In this paper, we show that the TOO scheme is susceptible to SKS attacks without malicious signers and WKS attacks in the presence of malicious signers.

The rest of the letter is organized as follows: In Sect. 2, we review the TOO scheme. In Sect. 3, we

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show that the TOO scheme is susceptible to key substitution attacks. And we end with concluding remarks in Sect. 4.

2 Review of the TOO Scheme

The Parameters: Let G_1, G_2 be two groups of the same prime order q . We view G_1 as an additive group and G_2 as a multiplicative group. P is a generator of G_1 . Assume that the Computational Diffie-Hellman Problem (CDHP) is intractable both in G_1 and G_2 .

The function $e: G_1 \times G_1 \rightarrow G_2$ is a pairing which satisfies the following properties:

- **Bilinear:** $e(aR, bQ) = [e(R, Q)]^{ab}$, for any $R, Q \in G_1$, and for any $a, b \in \mathbb{Z}_q$.

- **Nondegeneracy:** $e(P, P) \neq 1$.

- **Computable:**

There exists an efficient algorithm to compute $e(P, Q)$ for any $R, Q \in G_1$.

The TOO Scheme: With the above parameters $(G_1, G_2, q, P, e, H, n)$, the TOO scheme consists of three algorithms as follows:

- **Key Generate:**

- (1) Select $a_1, a_2, \dots, a_n \in \mathbb{Z}_q$.
- (2) Compute $V_1 = a_1 P, \dots, V_n = a_n P$

The public key is $\langle V_1, V_2, \dots, V_n \rangle$ and the corresponding private key is $\langle a_1, a_2, \dots, a_n \rangle$.

- **Signing :** Given a message $m \in \{0, 1\}^*$,

- (1) Compute $H(m) = (y_1, \dots, y_n)$, where $y_i \in \{0, 1\}$.
- (2) Compute $c \equiv (\sum_{i=1}^n a_i y_i)^{-1} \mod q$.
- (3) Compute $\sigma = cP$.

Then σ is a signature.

- **Verification:**

Given the public key $\langle V_1, V_2, \dots, V_n \rangle$, and the signature σ on m .

- (1) Firstly compute $H(m) = (y_1, \dots, y_n)$

- (2) Then compute $U = \sum_{i=1}^n y_i V_i$

Then σ is a valid signature on m under the public key $\langle V_1, V_2, \dots, V_n \rangle$ if and only if $e(U, \sigma) = e(P, P)$ holds.

3 Key Substitution Attacks on the TOO Scheme

In this section, we show that the TOO scheme is susceptible to SKS attacks without malicious signers and WKS attacks in the presence of malicious signers.

(1) Attacks on the TOO scheme without malicious signers.

We first observe the verification of the TOO scheme. During the verification of the TOO scheme, one need to compute $U = \sum_{i=1}^n y_i V_i$ from which we know that if $y_i = 0$ then the corresponding V_i does not contribute for the computation of U ; put another way, V_i is not used during the verification of TOO scheme.

In [4], the authors pointed out if a signature scheme has the property that some parts of the public key are not used during verification, then a general SKS attack can be mounted. Hence, the TOO scheme is susceptible to such general SKS attack. The idea of the general SKS attack is straightforward: modifying these public key parts which are not used during verification is enough to obtain a substitute public key.

To give an example, a SKS attack on the TOO scheme can be mounted as follows. Given message-signature pair (m, σ) under the public key $\langle V_1, V_2, \dots, V_n \rangle$, an adversary $\mathcal{A}$ computes $H(m) = (y_1, \dots, y_n)$ and searches out which y_i 's are zero. Without loss of generality, we assume that $y_1 = 0$, now $\mathcal{A}$ selects $a'_1 \in \mathbb{Z}_q$ and computes $V'_1 = a'_1 P$, $\mathcal{A}$ then sets $\langle V'_1, V_2, \dots, V_n \rangle$ as a

public key of another user. It is easy to verify that σ is also a valid signature on m under the public key $\langle V'_1, V_2, \dots, V_n \rangle$, for $e(y_1 V'_1 + \sum_{i=2}^n y_i V_i, \sigma) = e(\sum_{i=1}^n y_i V_i, \sigma) = e(P, P)$ holds.

(2) WKS Attacks on the TOO Scheme in the presence of malicious signers

With the presence of an malicious signer, the above described SKS attack on the TOO scheme can be converted into a WKS attack. To make a WKS attack work, $\mathcal{A}$ obtains the private key $\langle a_1, a_2, \dots, a_n \rangle$ corresponding to the public key $\langle V_1, V_2, \dots, V_n \rangle$ from the malicious signer, and hence knows the private key $\langle a'_1, a_2, \dots, a_n \rangle$ corresponding to the public key $\langle V'_1, V_2, \dots, V_n \rangle$ he chose.

5 Conclusion

In this paper, we show that the short signature scheme proposed by TSO et al. at ITNG'09 is susceptible to Strong-key substitution attacks without malicious signers and weak-key substitution attacks in the presence of malicious signers.

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